MIL-H-60648(MU)
17 March 1967
SUPERSEDING
FA-PD-MI-2521, Rev.1
8 May 1964

MILITARY SPECIFICATION
HOLDER, STRIP, LOADED, ASSEMBLY
M8, M9, M10, XM12, M19, or XM25 AIRCRAFT EJECTION SEAT CATAPULT

1. SCOPE

1.1 This specification covers Holder, Strip, Loaded Assemblies for use with Propellant Actuated Devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-17689 - Double Base Sheet Propellant, Type N5

Naval Ordnance

OS-11312 - Double Base Sheet Propellant, Type HE-N12

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Material
MIL-STD-1167 - Ammunition Data Card
MS-28741 - Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

DRAWINGS

Munitions Command

C8856870	- Carton, Packing, Ammunition for Loaded Strip Holder Assembly M10 or XM25
C8856871	- Box, Packing, Ammunition for Loaded Strip Holder Assembly, M10 or XM25
C8856872	- Carton, Packing, Ammunition for Loaded Strip Holder Assembly, M9
C8856873	- Box, Packing, Ammunition for Loaded Strip Holder Assembly, M9
C8856874	- Carton, Packing, Ammunition for Loaded Strip Holder Assembly, M8, XM12 or M19
C8856875	- Box, Packing, Ammunition for Loaded Strip Holder Assembly, M8, XM12, or M19
D8596873	- Holder, Strip, Assembly, Loaded
D8596927	- Holder, Strip, Loaded Assembly
F8596951	- Holder, Strip, Loaded Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement function shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Functional

3.2.1 The loaded Strip Assembly shall not fail to fire.

3.2.2 Ballistics.- The loaded Strip Assembly when fired with an acceptable Primary Igniter in an acceptable M8, M9, M10, XM12, M19, or XM25 Aircraft Ejection Seat Catapult (or an approved facsimile thereof) shall be capable of separating the launcher tube from the motor, actuating the piston slider valve and producing the ballistic requirements at the temperatures specified in Table I. Propelled mass shall be 400 pounds for M10 and XM25 Catapults and 350 pounds for M8, M9, XM12 and M19 Catapults.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

TABLE I

Temperature Fahrenheit(°F)	Catapult	Acceleration Maximum	Rate of Change of Acceleration max (see 6.2)	Launch Time max(See 6.3)	Acceleration Time Integral min(see 6.1)	Ignition Delay max(see 6.4)
-65 to -70°F	M8,XM12,M19	20.0g	300g/sec	240 millisec	0.8g.sec	60 millisec
-65 to -70°F	M9	20.0g	300g/sec	240 millisec	0.8g.sec	60 millisec
-65 to -70°F	M10, XM25	20.0g	300g/sec	240 millisec	0.8g.sec	60 millisec
65 to 75°F	M8,XM12,M19	20.0g	300g/sec	190 millisec	1.0g.sec	60 millisec
65 to 75°F	M9	20.0g	350g/sec	190 millisec	1.0g.sec	60 millisec
65 to 75°F	M10, XM25	20.0g	350g/sec	190 millisec	1.0g.sec	60 millisec
160 to 165°F	M8,XM12,M19	20.0g	300g/sec	170 millisec	1.2g.sec	60 millisec
160 to 165°F	M9	20.0g	450g/sec	170 millisec	1.2g.sec	60 millisec
160 to 165°F	M10, XM25	20.0g	450g/sec	170 millisec	1.2g.sec	60 millisec

3.2.2.1 Initiation.- The catapults shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of aircraft hose, size No. 4, standard MS-28741, at 70° + 5°F.

3.3 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, specifications, and the following:

3.3.1 Foreign matter.- The assembly shall not contain chips, dirt, grease, oil, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

4.1.1 Quality Assurance Terms and Definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as otherwise specified herein.

4.2 First Article Sample.

4.2.1 Initial Production Sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Inhibitant from not more than one lot.

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production.

4.2.1.1 Examination and Test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications, at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial Production Sample Failure.- Failure of the sample to meet the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection Provisions.

4.3.1 Lot.

4.3.1.1 Submission of Product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of Lot.- For the purpose of identification and shipment, the shipping lots of loaded strip holder assemblies shall consist of 1200 assemblies, maximum, exclusive of test samples, provided the homogeneity of the inspection lots has been retained as follows:

a. Components produced by a homogeneous process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

b. Propellant from not more than one lot.

c. Inhibitant from not more than one lot.

A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 -22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over - 43	6	8	11	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

Table III (Cont'd)

Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over - 48	8	11	17	20

4.3.2.1 Holder, Strip, Loaded, M8, M9, M10, M12, M19, and XM25, Assembly Drawing D8596873, F8596927 or F8596951, as applicable.

Defect	Method Of Inspection
CRITICAL	
1. Any part missing	Visual
2. Improper Assembly	Visual
3. Mixed holder strips	Visual
4. Metal defective	Visual
5. Marking missing, incorrect or illegible	Visual
6. Chamfer on propellant ends missing or incorrect	Visual
MAJOR	
101. Thickness of propellant strip	Gage
102. Inhibitor not dry, improperly cured or post cured	Manual & Visual
103. Size of propellant perforations incorrect	Gage
104. Number of propellant perforations incorrect	Visual
105. Silicone tape improperly applied	Visual
106. Inhibitor improperly applied	Visual
107. Spacing between propellant strips incorrect	Visual
108. Voids between propellant strip and inhibiting tape	Visual
109. Diagonal cutout missing	Visual
MINOR	
201. Radii missing or incorrect	Visual
202. Surface finish improper	Visual
203. Burr	Visual
204. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

4.3.2.2 Carton, Packing, Ammunition for Loaded Strip Holder Assembly, M8, M9, M10, XM12, M19, or XM25.- Drawing C8856870, C8856872, or C8856874, as applicable.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Visual
202. Quantity of contents incorrect	Visual
203. Separation or filling missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated layers	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.3 Box, Packing, Ammunition for Loaded Strip Holder Assembly, M8, M9, M10, XM12, M19 or XM25.- Drawing C8856871, C8856873 or C8856875, as applicable.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

4.3.2.3 MINOR (Cont'd)

Defect	Method Of Inspection
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection Testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Functional Tests.- The following tests shall be performed by the Government.

4.3.3.1.1 Ballistics.- Twenty-one loaded strip holder assemblies of the initial production sample and each production lot shall be temperature conditioned as specified in Table IV and in 4.4.1.1.1, and then tested in accordance with 4.4.1.1 to determine compliance with 3.2. Failure of any one of a test group to comply with the requirements of 3.2 shall be cause for rejection of the lot.

TABLE IV

Temperature (^{o}F)	Sample Sizes	
	Initial Production	Production Lot
-65 to -70	7	7
65 to 75	7	7
160 to 165	7	7

4.4 Test Methods and Procedures

4.4.1 Functional

4.4.1.1 Ballistics.- The sample shall be installed in an appropriate catapult or simulated equal. The head of the catapult shall be statically supported, and the test weight shall be attached to the mounting lugs of the launcher and allowed to move freely. The sample shall be ballistically tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned at the original temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

4.4.1.1.1 Temperature Conditioning.- A standard M8, M9, M10, XM12, M19 or XM25 Catapult, loaded with a dummy holder strip assembly and containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy holder strip to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Packing and marking shall be in accordance with the requirements of the applicable drawings.

5.2 Data Card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 "Acceleration Time" is defined as the area under the acceleration-time record bounded by the "launch time" limits.

6.2 "Rate of Change of Acceleration" is defined as the maximum positive slope of the acceleration-time record measured over any 30 milliseconds between the continuous rise and the maximum acceleration on the acceleration-time record.

6.3 "Launch Time" is defined as the time between the beginning of the continuous rise and the sudden drop on the acceleration-time record.

6.4 "Ignition Delay" is defined as the time that elapses between the firing of the catapult primary igniter (or the primary igniter in an approved test vehicle) and the beginning of the continuous rise on the acceleration-time record.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A086

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA- H1100
Philadelphia, Pa. 19137

FOLD

To detach this form cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:53Z
Check the source to verify that this is the current version before use.